



UNIVERSITY OF LEEDS

Formal Language & Finite Automata

Non-regularity

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All reading is from the "Introduction to the theory of computation" -
Micheal Sipser

- Pumping lemma for regular languages - Pages 77 - 82 (essential)

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When trying to design a deterministic finite state automata for it you encounter an issues. You would need an infinite number of states, one to represent the number of 0's that have been seen. However, deterministic finite state automata require there to be a finite number of states.

This is **not** a proof, rather an intuition. We will develop the tools we need to show that it is not regular next.

Pumping lemma for regular languages

Theorem (Pumping lemma for regular languages)

If A is a regular language, then there is a number p (the pumping length) where if s is any string in A of length at least p , then s may be divided into three parts, $s = xyz$, satisfying the following conditions:

1. *for each $i \geq 0$, $xy^iz \in A$,*
2. *$|y| > 0$, and*
3. *$|xy| \leq p$.*

All regular languages have this property.

Pumping lemma for regular languages

Proof.

Let $M = (Q, \Sigma, \delta, q_1, F)$ be a deterministic finite automaton that recognises language L and let $p = |Q|$. Let $s = s_1, \dots, s_n$ be a string in L of length n where $n \geq p$. Let $r_0, \dots, r_n$ be the sequence of states that M enters while processing s , so $r_i = \delta(r_{i-1}, s_i)$ for $1 \leq i \leq n$. This sequence of states has length $n + 1$, which is at least $p + 1$. By the pigeonhole principle in the first $p + 1$ elements of the sequence there must exist a repeated state. Let r_j be the first instance of the repeated state and let r_ℓ be the second instance of the repeated state. Because r_ℓ occurs among the first $p + 1$ elements in the sequence starting from r_0 , we have that $\ell \leq p + 1$. Let $x = s_1, \dots, s_j$, $y = s_{j+1}, \dots, s_\ell$ and $z = s_{\ell+1}, \dots, s_n$.

As x takes M from r_1 to r_j , y takes M from r_j to r_j , and z takes M from r_j to r_{n+1} (recall that r_{n+1} is an accepting state) then M accepts xyz .

The automaton M also accepts $xy^i z$ where $i \geq 0$. We know that $j \neq \ell$, so $|y| > 0$ and $\ell \leq p + 1$ so $|xy| \leq p$. □

Pumping lemma for regular languages

All regular languages have the pumping property.

If a language does not have the pumping property then it can't be regular.

Warning!!

There are non-regular languages which have the pumping property. This is not an **if and only if** statement.

$\{0^n 1^n \mid 0 \leq n\}$ is not regular

We proceed by contradiction. Assume that $\{0^n 1^n \mid n \geq 0\}$ is regular. Let p be the pumping length. We choose $s = 0^p 1^p$, clearly $s \in \{0^n 1^n \mid n \geq 0\}$ and as $|s| > p$ the pumping lemma says s can be divided into three sections x, y and z such that $s = xyz$, where for any $i \geq 0$ the string $xy^i z \in \{0^n 1^n \mid n \geq 0\}$, $|y| > 0$ and $|xy| \leq p$. Observe the following;

- Since $|xy| \leq p$ we know that $x = 0^k$, $y = 0^\ell$ and $z = 0^{p-k-\ell} 1^p$ for some $k, \ell \in \mathbb{N}$.
- Consider $xyyz$ and observe that $xyyz = 0^{p+\ell} 1^p$.
- Since $|y| > 0$ we know that $\ell > 0$ and hence $xyyz \notin \{0^n 1^n \mid 0 \leq n\}$, a contradiction.